

State Diagram – consisting of states, transitions, events, and activities -- model the dynamic nature of a system. Specifically it define different states of an object during its lifetime and these states are changed by events.

State – often used by chart interchangeably—is defined as a condition in which an object exists and it changes when some event is triggered

Event -- an occurrence of a stimulus that can trigger a state transition.

Transition—a relationship between two states indicating that an object in the first state will perform certain actions and enter the second state when a specified event occurs and specified conditions are satisfied.

Action -- an executable atomic computation that results in a change in the state of the model or the return of a value.

Notations

1. initial state:



2. state:



2. composite state:



3. final state:



4. self transition:



5. transition:



1. A state diagram of the microwave oven heating system.

Description—Usually people turn the microwave oven on to heat food with a timer. The heater will not stop when the timer is in use. Otherwise, the heater is idle because the microwave oven is still on.